

Institute of Actuaries of India

Subject CP1 – Actuarial Practice (Paper A)

MAY 2023 EXAMINATION

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

1. To cover the research and development costs for new technology in green energy space
 - a. which may be significant and long-term, occurring before any revenue is generated
2. to cope with mismatches in costs and revenues - in particular, before revenues are received, there will often be massive capital outlays, including:
 - a. the purchase of raw materials
 - b. marketing costs
 - c. manufacture of equipment's like solar panels which can harness/convert the green energy into profitable product
3. Cushion against fluctuating trade volumes
4. To meet the costs arising on unexpected events, eg:
 - a. a cancellation of a major order, eg an airline cancels an order for a new fleet due
 - b. to a reduction in demand for green energy
 - c. any court awards for liability in the event of any accident or adverse impact on the environment from the use of company's technology
 - d. adverse currency movement leading to higher than expected research and development cost
5. to take advantage of any opportunities:
 - a. expanding into new markets
 - b. merger and acquisitions of companies in the value chain
 - c. undertaking projects which have the potential to create more profitable and efficient business

start-up capital ,if the company aims to expand its business to meet infrastructural and regulatory costs.

[6 Marks]

Solution 2:

The 3 principles of insurance and pensions are:

1. The existence of insurable interest:
 - a. An insurance contract is only valid if the person taking out the contract has s financial interest in the insured event
 - b. This is primarily to prevent moral hazard.
 - c. Individuals are assumed to have unlimited financial interest in their own life and the lives of sources and dependent children.
 - d. Other financial interests are limited in amount to prevent over insurance
2. Pre-funding:
 - a. The key principle of insurance and pensions is that individuals or corporate bodies put money aside in advance of the occurrence of an uncertain risk event. The uncertainty might relate to:
 - a. whether the event will happen at all, such as the risk of fire or flood the
 - b. timing of a certain risk event, such as life expectancy
 - c. the cost of an event that is certainly going to occur.

The key issue for the individual or corporation is how much money is needed to provide a given level of benefit with the desired probability. This will depend on:

- d. the probability of the risk event occurring the amount that the risk event will cost, and
- e. the return that can be earned on the pre-funded money before the risk event occurs.

The individual will also have a risk tolerance - how comfortable they are with the probability of their desired outcome not being achieved.

3. Pooling of risk:
 - a. Pooling of finances helps to protect the individuals against some of the uncertainties the may exist in the cost of financing the benefits. It may also lead to more cost effective provisions than if each individual made their own financial provisions.

[6 Marks]

Solution 3:

- i) OPS is a Defined benefit scheme whereas NPS is a Defined contribution scheme.

A defined benefit scheme is one where the scheme rules define the benefits independently of the contributions payable, and benefits are not directly related to the investments of the scheme.

A defined contribution scheme is one providing benefits where the amount of an individual member's benefits depends on the contributions paid into the scheme in respect of that member, increased by the investment return earned on those contributions.

[2]

- ii) Government should prefer to continue New Pension Scheme to realise following objectives:

- To incentivise private savings and hence reduce the burden on government in future.
- The Scheme may also be a means of meeting political end, i.e, make people self reliant, build up assets and have more sense of ownership over public property and hence a more harmonious society.
- It is possible that the defined benefits may be inadequate for the majority of the population and moreover this benefit is not indexed/inflation proof. Increasing the same in future or indexing the same might be huge burden for the government.
- To make younger population feel the importance of saving for retirement from an early age.
- To give an opportunity to an individual to plan for old age in a tax efficient and inexpensive way (as the fees will be set by the Government)
- Because of the tax breaks and the availability of minimum pension the Government can still claim “availability of reasonable social security”.
- Due to the tax breaks there will be loss of revenue (at least initially) to Government and savings take a long time to come through –
But long term savings etc could be worth it if they can cover the initial shortfall.
Moreover, as this is a developing country increasing pension savings will enable investments in long term projects. This is a source of private capital and will reduce need for government spending or borrowing. Extra growth created could feedback to higher tax revenues.
- Pension savings can be expected to be more long term than any other form of savings. Note that the NPS has to mandatorily invest only in domestic assets.
- To inculcate financial discipline amongst citizens to provide for retirement benefits.
- To improve depth of financial markets as NPS funds would be invested in securities market as per choice of NPS subscribers.

[5]

- iii) Proposal to restore OPS shall have implication on various stakeholder:

- State Government: Pensions are paid for by the government. As a result, given same sources of revenues, state government will find itself financially constrained if OPS benefits are far generous than NPS benefits.
- Employees & beneficiaries: Risk of level of pension benefits available upon retirement is assumed by employees in NPS. Once OPS is restored, employees/beneficiaries shall be relieved of this risk. There may be change in present salary structure due to transition from NPS to OPS.
- Tax Payers: Tax payers face the risk of higher taxes and greater tax scrutiny as state government shall be under constant pressure to shore up revenues.
- Administration: State government shall have to establish department that would be entrusted responsibility of administering OPS benefits. Regular actuarial valuation of liability, investment/management of scheme assets, establishing eligibility for benefit payout and actual benefit payout are few tasks that shall need to be assumed by government.

- Other Benefit providers: NPS fund managers would find their AUM growth declining as now NPS fund shall not augment as anticipated once OPS is implemented. State government may also call upon fund manager to entrust funds built up so far and transfer them under state government custody/control.
- Insurers: Under NPS arrangement, upon retirement employees are required to purchase annuity from empanelled insurers. Once OPS is implement, unless state government too purchases annuity there shall be no annuity new business for state employee pension benefits.

[4]

[11 Marks]

Solution 4:

- i) Implied cost of elongated warranty shall be calculated as:
Expected claim frequency * expected cost per claim

Since such complimentary warranty is offered for first time, the company will not have any past data to estimate claim frequency and cost per claim that arises during such elongated period.

The nature of risk is heterogeneous (paint work & body work erosion) and therefore it may be difficult to arrive at standard implied cost.

[2]

- ii) To determine provision for unexpired warranty period, we would need following information:

- Make of the car
- Mileage (both historical and annual)
- Car model
- Car colour if the cost of repairs vary with the colour
- Protective measures like Teflon coating, if any undertaken to protect car colour
- Intended use of car i.e., personal use/on hire
- Car ownership status i.e., first owner or subsequent

[3]

- iii) Aspects to be considered while designing product:

- Scope, period and extent of coverage offered under warranty
- Coverage exclusions that can be introduced yet maintaining product attractiveness
- Frequency of claim experience review and agreeing to reprice when adverse experience emerges
- Veracity of claim underwriting – Own staff vs. third party service providers
- Information collected from car owner at the time of car sale for e.g., past traffic law violations, usual place where car shall be parked, purpose of car usage i.e., personal use or commercial use, self-driven or hired driver driven etc.
- Regulatory guidance on product design
- Cooling period between two claims
- Who would perform repairs under warranty? – Authorised service station or independent mechanic/garages or own service centres

Aspects to be considered while Pricing product:

- Past claim experience/burning cost
- Capital and provisioning requirements
- Profit target
- Auto co.'s budget
- Rate guarantee expected
- Overhead and direct costs to be loaded in price

[5]

- iv) Key risks that GI company would face are:
- underwriting risk – arising in relation to the underwriting approach taken
 - insurance risk – arising from the uncertainties relating to claim rates and amounts
 - financing risk – arising in relation to the financing of claims cost and expenses
 - exposure risk – arising in relation to the amount of business sold or retained, or to its concentration or lack of diversification
 - Operational risk – Operational risk refers to the risk of loss resulting from inadequate or failed internal processes, people and systems or from external events.
 - External risk - External risk arises from external events, such as
 - storm, fire, flood, or terrorist attack

[3]

[13 Marks]**Solution 5:**

- i) High liquidity policy and close to nil risk free interest rate pursued by the Central bank might have caused large influx of deposits with Banks.

Banks in turn invest the funds in various assets like loans to borrowers, treasury bonds, corporate bonds and other securities that provide fixed return like mortgage bond securities.

All the banks have liabilities that have short duration while their assets have longer duration. Banking business works on the premise that only small fraction of depositors would demand their money and they can be settled by holding sufficient liquid assets.

Where banks resort to asset allocation that is more skewed towards longer duration or greater duration, it is accepting higher risk of asset liability mismatch.

Consequent to central bank taking contractionary monetary policy, rise in yields caused fall in market value of bonds. Fall is more pronounced towards higher asset duration.

Thus, if market value of assets fall, bank would suffer mark to market loss on assets held by it. These losses are realised in case bank sells them.

To aggravate the situation, with rise in yields/interest rates, depositors would be keen to take away their own funds/deposits as money supply is reduced and credit is available at higher cost.

Banks would be forced to liquidate its assets where depositors ask for their money. In other words, banks are forced to book their losses on bonds portfolio.

Banks are also required to hold certain level of capital to support its business. Substantial loss would drop the available capital and bank might appear under capitalised.

Fortune of large banks are intertwined with other banks. Whenever any bank faces financial difficulties it creates ripple effect across banking system. Thus, causing domino effect/systemic issue.

[6]

- ii) A bank with home loan book would receive a stream of cash inflows in form of interest and principal repayment. These cash inflows shall arise over tenure of loans and are usually secured against mortgage of home.

Bank in need of immediate funds can consider securitisation of home loan book. Securitisation involves transferring loan assets to a special purpose vehicle (SPV). Special purpose vehicle shall raise funds from investors by issuing shares/unit of SPV.

SPV shall in turn pay consideration for home loan book to original lender. Now SPV shall receive all cash inflows from home loan book and distribute them to unitholders.

Securitisation shall be successful when expected return for unit holders is commensurate with the risks assumed by investing in securitised assets. In other words, holders of securitised asset would expect rate of return equivalent or may be better than original lender.

Securitisation would immediately result in liquidation of assets for bank. Now bank would hold more cash and in better position to meet depositors demand. However, securitisation would also mean that Bank is foregoing net interest margin that it would have earned by holding the loan book until its runoff.

[4]

- iii) Bank can raise capital through issuing subordinated debt in the capital markets. The main aim of subordinated debt is to generate additional capital that improves the free capital position, as the debt does not need to be included as a liability in the assessment of solvency.

The subordinated debt can only pay interest if regulatory solvency capital requirements will continue to be met after the interest is paid or, in some countries, if authorised by the regulator. Repayment of capital can only be made if, after repayment, regulatory solvency capital requirements continue to be met and if they are authorised by the regulator.

In the event of wind-up, the subordinated debt in all cases ranks behind bank's liabilities (other than shareholder's capital).

So again, this method of raising capital has increased the assets of the provider (by the amount of the debt issued) but, because the repayments rank behind the bank's liabilities (i.e., are contingent on those liabilities being met), it does not increase the liabilities. Therefore, once again the bank's available capital position may be improved.

Bank would find it difficult to raise capital through subordinated debt if there is any recent incident of subordinated debt written down. Bank that is sound would find many takers of subordinate debt at low cost but Bank that is facing financial difficulties would have to be prepared to bear higher cost of subordinated debt. Latter bank would be in more need of capital but cost might become prohibitive.

[4]

[14 Marks]

Solution 6:

- i) A claim is payable on this contract when the policyholder is deemed to have reached a specified level of disability, for example the policyholder may be unable to perform a specified number of 'activities of daily living' (ADLs).

The different levels of care will differ between one contract and another, but typically may include the following (in increasing order of cost):

- a. cost of care in own home
- b. cost of being cared for in a residential (but non-nursing) home
- c. cost of being cared for in a residential nursing home.

The contract can be paid for by single or regular premiums, and all types of benefit structures (without-profit, with-profit, unit-linked and investment linked) are possible. Any regular premiums would cease from the point at which claims begin to be paid (if not at some earlier date).

The contract can be used to help provide financial security against the risk of needing either home or nursing-home care as an elderly person, ie post-retirement. The contract could pay for all the

costs of care throughout the remainder of life (an indemnity contract), or could provide a cash lump sum, or an annuity, to contribute towards the costs of care.

[4]

ii)

1. Key aspects affecting setting the assumptions are:
 - a. The use to which the model will be put
 - b. The financial significance of the assumptions
 - c. Consistency between the assumptions – compared to current product or other similar products
 - d. Legislative and regulatory requirements – tax, regulations
 - e. The need of the stakeholders
2. How to gather the data:
 - a. Does the company in the past or currently writing any long term care product? If yes, then assumptions underlying the product can be used.
 - b. If no, then alternative source of data is required like:
 - i. Reinsurance, Will the reinsurer participate in this product and what will be the reinsurance cost that needs to be allowed for?
 - ii. Standard table:
 1. important to understand if the standard tables are aligned to the target customer demographics
 2. To assess what % age of such standard table will be more suitable for the company to reflect its expected experience
3. These assumptions need to be reviewed considering the business strategy that the company has for growth of this segment.
 - a. Factors to be considered?
 - i. Is company planning to use any new distribution?
 - ii. Is company planning to sell into different geographies than it is currently selling?
 - iii. Is the expectation of margin still the same?
 - iv. Is any change in UW expected, as this will impact the mortality and longevity assumptions
 - v. Incentive like customer discount, higher commissions, lower acquisition cost, higher expense towards marketing etc. needs to be allowed for, especially if company want to capture the market in a short span of time
4. Can company tie up with diagnostic centres and for initial or regular check-ups of its customers?
 - a. This will ensure that the data used for UW is credible and cost of acquisition and regular check-up happens at a reasonable rates.
 - b. Whether company can have these tie ups will impact the acquisition and maintenance expense assumptions
5. Sensitivity testing and scenario analysis, especially to assess the impact of longevity risk and what loading is needed to maintain target profit margins
6. ALM consideration and how will it impact the interest rate assumptions
7. Capital requirements and cost of capital assumptions
8. Assumptions and MAD that is needed while reserving as that will have an impact on NB strain and margins of the product]

[8]

iii)

1. If the company writes only/predominantly annuity business, then longevity risk is fundamentally a systematic risk. If the mortality rates are falling for the population as a whole, this risk cannot be diversified away.
2. If the longevity rates vary for certain subsets of the population (certain socio economic profiles) then the company may be able to diversify its exposure by either avoiding such socio

economic profiles or by targeting a wider range of population with varying socio economic profile

3. It can reduce volatility and hence diversify the risk by selling large number of policies, thereby benefitting from law of large numbers
4. The company can diversify the risk by writing term or whole life products as improving mortality will have a beneficial impact of such products which will negate the impact of longevity risk on long term care product
5. Company can also diversify longevity risk against other risks like market, operational risk, thereby reducing the overall capital requirement of the company.

[3]

[15 Marks]**Solution 7:**

- i) One may invest in Gold as an hedge against inflation. Gold is considered as a safe haven and not correlated with the volatility of financial instruments. Gold can be preferred over cash in medium to longer term where real value of gold is maintained despite inflation.

[2]

- ii) Physical gold involves buying real gold bar/coins and/or jewellery.

Investing in physical gold has following merits:

- Better and Instant liquidity in physical gold vs. any other instrument with underlying gold
- Feel good factor for people fond of jewellery and pleasing near and dear ones

Demerits of investing in physical gold:

- High cost of holding in terms of security/bank locker rent/peace of mind etc.
- High dealing cost in form of GST on purchase of physical gold
- Relatively high making charges/wastage in ornamenting jewellery that implies immediate loss of capital
- Risk of buying poor purity gold
- Usually lay investors are not savvy in ascertaining real/realisable value of jewellery while buying jewellery
- Need to maintain paper trail/documents to establish cost/sale consideration for tax purposes.
- Unless bought in smaller quantities, Physical gold is not granular to be liquidated in parts as and when required.
- Capital invested in gold is not available as a factor of production/to further economic activity.

Gold ETF is a mutual fund scheme wherein fund manager collects money from investor and purchases real gold. In exchange, investor is allotted units of the mutual fund scheme that can be traded on stock exchange. NAV of Gold ETF are regularly computed and published. In this way investor gets exposure to Gold without facing associated hassles of managing it.

Merits of Gold ETF:

- No hassles of managing physical gold like security, adulteration etc.
- Documentation of sale & purchase is easily available/retrievable from depository/asset management company. Simple profit/loss reports are available that can be used for tax compliance
- Professional fund manager are expected to be more vigilant/informed while making purchase/sale decisions so information asymmetry faced by individual while dealing in real gold is not present in ETF.
- ETF units can be bought & sold multiple of single unit representing minute quantity of underlying gold.

Demerits of Gold ETF:

- Additional cost of fund management fee required to be borne by investor
- ETF units are not traded often and there are liquidity problems when we wish to buy/sell
- Traded ETF unit price may not closely reflect underlying NAV. Hence, returns generated may suffer from tracking error.
- Instant liquidity is not available as units can be redeemed/traded during working hours of working day and money is realised with some lag.

Sovereign Gold Bonds (SGB) are fixed interest debt security issued by Government. Each bond represent one gram of gold. SGB entitle coupon payments are regular interval and are redeemed at prevailing gold index linked price after 8 years of issue. Coupon payments are taxable as normal income while capital gain on maturity is exempt from tax. SGB are traded on stock exchanges.

Merits of SGB:

- Perfect tracking of gold returns in addition to regular coupon yield. Minor residual tracking error expected between cutoff date of ascertaining gold price index and actual redemption.
- No fund management cost whatsoever
- No hassles of managing physical gold like security, adulteration etc.
- Superior returns as compared to other gold instruments as coupon payments and tax free capital gains improve post tax returns.
- Earlier liquidity available in secondary market.
- Can be traded in smaller unit of 1 gram gold bond.
- Documentation of sale & purchase is easily available/retrievable from depository/broker. Simple profit/loss reports are available that can be used for tax compliance
- Conservation of foreign currency that would otherwise be spent in purchasing imported real gold
- Funds raised against SGB are invested by Government in economy for capital/revenue expenditure thereby causing multiplier effect/better economic growth.

Demerits of SGB:

- Instant liquidity is not available as SGB can be traded during working hours of working day and money is realised
- with some lag.
- Exit before maturity might lead to realising less than fair value as SGB are thinly traded and price discovery is inefficient.

[10]

iii) Investor may take exposure to gold following other ways:

- Equity shares of Jeweller/Gold lenders: Though the returns are loosely correlated to gold price
- Digital Gold: This involves purchase of gold from third parties/fintech entities who hold custody of gold on buyer's behalf. Buyer may choose to sale gold back or take delivery of gold
- Commodity derivatives in Gold: Taking position of gold futures on commodity exchanges. However, since futures are meant for short terms there is need to rollover contract and incur dealing cost. Position in commodity future may turnout to be very risky if speculated taking high leverage.

[3]

[15 Marks]**Solution 8:****i)**

1. Faster approvals of product design so that companies can commence business

2. Allowing frequent review of premiums so that companies can align the emerging experience
3. Lobby with the government for any tax benefits on such products
4. Regulatory support in reduced restriction in respect to expense of management targets
5. Asking reinsurers to also partner with companies to share the risks to help increase penetration
6. Providing industry/geographic level data related to customer demographic profiles for company to assess the assumptions that it should be taking while pricing the products
7. Ensuring that all companies as part of the industry are proportionally tasked with the penetration objective and its not only that the big players are tasked with higher targets which may impact their business.

[3]

ii)

1. Which are the geographies that the company is expected to drive penetration drive
2. What is the expectation or targets as set by the regulator? Are these target sustainable or not.
3. What is the period over which these targets needs to be achieved? How the end target can be split into short term targets?
4. What is the customer demographic of the geographies that have been assigned to the company
5. What would be the capital requirement of this initiative?
6. Can company's current solvency level adequate to achieve its own plans as well as this new target?
 - a. If not, what are the source of capital that the company can look at?
7. Review of its product suite, can it use any of the current products that it can use to drive the penetration or does it need a new product all together?
 - a. Cost of setting this product?
 - b. Review of current assumptions while setting up a new product
8. What changes are required in distribution strategy?
9. If the company is listed, then it needs to be assessed what impact will driving this initiative will have on the profitability metrics and valuation of the company and hence, on the share price of the company. What actions are needed to ensure that impact on share price is limited (if adverse)
10. What support is expected from the government or regulator in respect to taxes, expense allowance etc. all of which will impact the pricing as well as the profitability of this initiative.

[6]

iii)

1. Current products may t be complex to be sold in those cities.
2. Customer needs in those cities can be different hence the current products are not feasible
3. UW requirement of current products may not be possible to be met in those cities, especially medical UW
4. A simple product with reduced UW may be required and hence a new product would be needed
5. Assumptions mainly mortality and expense assumptions underlying the current products will not be aligned with the expected mortality or expense:
 - a. Mortality expectations would be higher than currently assumed, hence current pricing will not be sustainable
 - b. Cost of acquisition in these cities would be higher than what is current, hence current price will not be sustainable
6. Significant deviation of actual assumptions to expected can have a capital impact
7. There may be tax advantages being given by the regulator which the current product features may not be aligned to, hence a new product design may be needed.

8. A new distribution set up may be needed to grow into these cities and hence a simple product would help in ensuring good sales practices.

[5]

iv)

1. A simple product design would be more acceptable.
2. Pricing assumptions regarding mortality, persistency and expenses should allow for an adverse experience as compared to its current experience.
 - a. For mortality assumption, company can use standard table or competition data or work with reinsurer to price the product.
3. Look to transfer the risk through reinsurance
4. Review of UW requirements, can be make it simple (especially medical UW) if a simple design product is sold
5. No sale in places tagged as negative area for sale
6. Limiting the SA that can be sold in these cities
7. Limited or no wavier (financial or medical) to any distribution channel while getting new business.
8. Close monitoring of claims experience and continuous feedback to UW teams on the emerging trend
9. Close monitoring by the risk team of the experience against the risk appetite as defined and taking business actions when the appetite is about or is breached.
10. Choosing not to sell term products in phase 1 till the time company would have got required experience to appropriately price a term product.
11. Higher due diligence at customer onboarding stage like visit to customer's place or asking customer to come to the branch for completion of formalities etc.
12. Reserving assumptions can be more prudent or allow for higher MAD to allow for expected adverse experience as compared to current products.

[6]

[20 Marks]
